

1 / 20 | Ages 9-12 | 20 minutes

The river crossing plan

What we are practising

Plan forward and backward moves while checking every intermediate state.

What you need

Paper / Pencil / Four large paper labels

Draw the riverbanks and write positions instead of moving cards.

Before you begin

Use a farmer, fox, chicken and grain. The boat carries the farmer plus at most one other item.

This is a paper logic story only; do not approach water or involve real animals.

Try it together

1. Start all four on the left bank; only the farmer operates the boat.
2. Do not leave fox with chicken, or chicken with grain, on a bank without the farmer.
3. Try crossings and record both banks after every move.
4. Continue until everyone reaches the right bank safely within the puzzle rules.

Talk, draw or show your idea

Why might carrying something back help the plan succeed?

Space to explore

2 / 20 | Ages 9-12 | 15 minutes

Only one label is true

What we are practising

Test candidate solutions against an exact truth condition.

What you need

Paper / Pencil

Draw labelled circles instead of boxes.

Before you begin

Draw red, blue and green boxes; exactly one contains a prize and exactly one label is true.

Use names as well as colours; no hidden real prize or reward is needed.

Try it together

1. Write red label: The prize is in blue. Blue label: The prize is not here. Green label: The prize is not in red.
2. Assume the prize is in red and mark all three labels true or false.
3. Repeat assuming blue, then green.
4. Choose the location producing exactly one true label.

Talk, draw or show your idea

Why must you test every label for each possible location?

Space to explore

3 / 20 | Ages 9-12 | 20 minutes

The shortest delivery route

What we are practising

Compare route totals instead of judging by drawing distance.

What you need

Paper / Pencil

Write the possible connections in a table instead of drawing a network.

Before you begin

Draw points A, B, C, D, E and connect $AB=2$, $AC=5$, $BC=1$, $BD=4$, $CD=1$, $CE=6$, $DE=3$. All links work both ways.

All routes are fictional; do not use the drawing for real travel.

Try it together

1. Find two routes from A to E and add their labelled costs.
2. Compare reaching C directly with reaching it through B.
3. Use the cheaper intermediate choices to find a better route.
4. Check the final total against the other routes you tried.

Talk, draw or show your idea

Can a route with more links still have a lower total cost?

Space to explore

4 / 20 | Ages 9-12 | 15 minutes

Crack the four-digit order

What we are practising

Use adjacency and ordering constraints to eliminate permutations.

What you need

Paper / Pencil

Use large handwritten cards for digits 1, 2, 3 and 4.

Before you begin

The code uses 1, 2, 3 and 4 exactly once, read from left to right.

Use an invented puzzle code, never a real password or device PIN.

Try it together

1. Write the clues: 1 is immediately before 3; 4 is before 2; 2 is not last.
2. Treat 1 and 3 as an ordered block and try possible placements.
3. Reject any order that breaks one clue.
4. Check the remaining code against all three clues.

Talk, draw or show your idea

How does treating 13 as a block reduce the search?

Space to explore

5 / 20 | Ages 9-12 | 20 minutes

Thirteen-token strategy

What we are practising

Reason backward from a target and maintain a useful total.

What you need

13 large paper tokens / Paper / Pencil

Draw 13 circles and cross them out.

Before you begin

Two players alternate taking 1, 2 or 3 tokens; whoever takes the last token wins.

Keep competition friendly and use large tokens around younger children.

Try it together

1. Play once from 13 tokens and record each move.
2. Study what happens when a player starts a turn with 4 tokens.
3. Look for replies that make the two players' moves total 4.
4. Start again and test taking 1 first, then completing each pair to 4.

Talk, draw or show your idea

Why is leaving 4 tokens useful just before the other player's turn?

Space to explore

6 / 20 | Ages 9-12 | 15 minutes

Every fruit label is wrong

What we are practising

Propagate deductions through incorrect labels.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Three boxes hold only apples, only bananas, or both; labels are Apples, Bananas, Mixed, and every label is wrong.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Three boxes hold only apples, only bananas, or both; labels are Apples, Bananas, Mixed, and every label is wrong.
2. Ask: You see an apple from the box labelled Mixed. What does each box actually contain?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Three boxes hold only apples, only bananas, or both; labels are Apples, Bananas, Mixed, and every label is wrong. You see an apple from the box labelled Mixed. What does each box actually contain?

Space to explore

7 / 20 | Ages 9-12 | 15 minutes

Two speakers, fixed habits

What we are practising

Test truth assignments for consistency.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Each speaker either always tells truth or always lies. A says “B lies.” B says “Exactly one of us tells truth.”

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Each speaker either always tells truth or always lies. A says “B lies.” B says “Exactly one of us tells truth.”
2. Ask: Who tells truth and who lies?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Each speaker either always tells truth or always lies. A says “B lies.” B says “Exactly one of us tells truth.” Who tells truth and who lies?

Space to explore

8 / 20 | Ages 9-12 | 10 minutes

The earliest project finish

What we are practising

Find a finish time from dependencies and parallel work.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Two people can work in parallel: cutting takes 3 minutes, folding 2; assembly takes 4 after both, then drying takes 5.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Two people can work in parallel: cutting takes 3 minutes, folding 2; assembly takes 4 after both, then drying takes 5.
2. Ask: Starting both first tasks at minute zero, what is the earliest finish time in this paper schedule?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Two people can work in parallel: cutting takes 3 minutes, folding 2; assembly takes 4 after both, then drying takes 5. Starting both first tasks at minute zero, what is the earliest finish time in this paper schedule?

Space to explore

9 / 20 | Ages 9-12 | 15 minutes

Can three lights all turn on?

What we are practising

Identify an invariant that proves impossibility.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Three drawn lights start off. Button A flips lights 1 and 2; B flips 2 and 3; C flips 1 and 3.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Three drawn lights start off. Button A flips lights 1 and 2; B flips 2 and 3; C flips 1 and 3.
2. Ask: Can any sequence of presses leave all three lights on? A flip changes on to off or off to on.
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Three drawn lights start off. Button A flips lights 1 and 2; B flips 2 and 3; C flips 1 and 3. Can any sequence of presses leave all three lights on? A flip changes on to off or off to on.

Space to explore

10 / 20 | Ages 9-12 | 15 minutes

A four-by-four number square

What we are practising

Combine intersecting constraints in a grid.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw rows 1 _ _ 4 / _ 4 1 _ / 2 _ 4 _ / _ 3 _ 1, read left to right.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw rows 1 _ _ 4 / _ 4 1 _ / 2 _ 4 _ / _ 3 _ 1, read left to right.
2. Ask: Fill 1-4 once in every row, column and two-by-two corner block. What is the completed grid?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw rows 1 _ _ 4 / _ 4 1 _ / 2 _ 4 _ / _ 3 _ 1, read left to right. Fill 1-4 once in every row, column and two-by-two corner block. What is the completed grid?

Space to explore

11 / 20 | Ages 9-12 | 10 minutes

Walk every bridge once

What we are practising

Distinguish revisiting places from reusing connections.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw undirected bridges AB, BC, CA and AD; there are no others.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw undirected bridges AB, BC, CA and AD; there are no others.
2. Ask: Can you start at D and cross every bridge exactly once? Give a route.
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw undirected bridges AB, BC, CA and AD; there are no others. Can you start at D and cross every bridge exactly once? Give a route.

Space to explore

12 / 20 | Ages 9-12 | 15 minutes

Three-digit clue laboratory

What we are practising

Reconcile exact membership and position clues.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A code uses three different digits from 1-5, read left to right. For 123: one digit present, none placed right; 452: two present, one right; 314: two present, one right.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A code uses three different digits from 1-5, read left to right. For 123: one digit present, none placed right; 452: two present, one right; 314: two present, one right.
2. Ask: What code satisfies all three clues? Counts describe exactly how many digits occur and are correctly placed.
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A code uses three different digits from 1-5, read left to right. For 123: one digit present, none placed right; 452: two present, one right; 314: two present, one right. What code satisfies all three clues? Counts describe exactly how many digits occur and are correctly placed.

Space to explore

13 / 20 | Ages 9-12 | 10 minutes

The club overlap

What we are practising

Correct double-counting in overlapping groups.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Twelve children each join art, science, or both; 8 join art and 7 join science.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Twelve children each join art, science, or both; 8 join art and 7 join science.
2. Ask: How many join both clubs?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Twelve children each join art, science, or both; 8 join art and 7 join science. How many join both clubs?

Space to explore

14 / 20 | Ages 9-12 | 10 minutes

What follows from some?

What we are practising

Separate a valid inference from an unsupported universal claim.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

All circles in a collection are smooth. Some red pieces are circles.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. All circles in a collection are smooth. Some red pieces are circles.
2. Ask: Must some red pieces be smooth? Must every red piece be smooth?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

All circles in a collection are smooth. Some red pieces are circles. Must some red pieces be smooth?
Must every red piece be smooth?

Space to explore

15 / 20 | Ages 9-12 | 10 minutes

Rotate the round-table plan

What we are practising

Recognise a relationship preserved by rotation.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Four seats around a table hold A,B,C,D clockwise; everyone moves one seat clockwise.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Four seats around a table hold A,B,C,D clockwise; everyone moves one seat clockwise.
2. Ask: Who remains opposite A after the move?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Four seats around a table hold A,B,C,D clockwise; everyone moves one seat clockwise. Who remains opposite A after the move?

Space to explore

16 / 20 | Ages 9-12 | 15 minutes

Nine choices and yes-or-no clues

What we are practising

Bound a search using binary decisions.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A hidden item is one of nine labelled pictures. Each question must have only yes or no as answers.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A hidden item is one of nine labelled pictures. Each question must have only yes or no as answers.
2. Ask: Can three questions always identify it, even when later questions depend on earlier answers?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A hidden item is one of nine labelled pictures. Each question must have only yes or no as answers. Can three questions always identify it, even when later questions depend on earlier answers?

Space to explore

17 / 20 | Ages 9-12 | 10 minutes

How many knockout matches?

What we are practising

Count necessary changes to reach a final state.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Eight teams enter a single-elimination tournament; each match eliminates exactly one team and there are no draws.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Eight teams enter a single-elimination tournament; each match eliminates exactly one team and there are no draws.
2. Ask: How many matches are needed to leave one champion?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Eight teams enter a single-elimination tournament; each match eliminates exactly one team and there are no draws. How many matches are needed to leave one champion?

Space to explore

18 / 20 | Ages 9-12 | 10 minutes

A five-square picture code

What we are practising

Decode run-length constraints.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw five cells. From left to right, filled runs must be length 2 then length 1, with at least one empty cell between; the first cell is empty.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw five cells. From left to right, filled runs must be length 2 then length 1, with at least one empty cell between; the first cell is empty.
2. Ask: Which cells are filled, numbering from the left?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw five cells. From left to right, filled runs must be length 2 then length 1, with at least one empty cell between; the first cell is empty. Which cells are filled, numbering from the left?

Space to explore

19 / 20 | Ages 9-12 | 15 minutes

The damaged checkerboard

What we are practising

Use colour counts to prove a tiling impossible.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a four-by-four alternating black/white board; remove top-left and bottom-right corners, which share a colour.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a four-by-four alternating black/white board; remove top-left and bottom-right corners, which share a colour.
2. Ask: Can seven dominoes, each covering two side-adjacent squares, cover the remaining board?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a four-by-four alternating black/white board; remove top-left and bottom-right corners, which share a colour. Can seven dominoes, each covering two side-adjacent squares, cover the remaining board?

Space to explore

20 / 20 | Ages 9-12 | 10 minutes

Everyone meets once

What we are practising

Avoid duplicate counting of unordered pairs.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Six people attend a pretend meeting; each pair has exactly one conversation together.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Six people attend a pretend meeting; each pair has exactly one conversation together.
2. Ask: How many distinct pair conversations are there?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Six people attend a pretend meeting; each pair has exactly one conversation together. How many distinct pair conversations are there?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. The river crossing plan

A little hint

Take the chicken first; later you will need to bring it back once.

Answer / what to notice

One solution: take chicken across; return alone; take fox across; bring chicken back; take grain across; return alone; take chicken across.

Bringing the chicken back prevents either forbidden pair from being left together without the farmer.

Make it easier

Provide the first two crossings and ask for the next one.

Take it further

Find the alternative solution that transports grain before fox.

Another day, away from the screen

Replace the characters with invented robots that cannot be left in certain pairs.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Only one label is true

A little hint

If the prize is in red, the blue label says something true.

Answer / what to notice

The prize is in red: red label false, blue true, green false. Blue or green each makes two labels true.

Only the red location satisfies both one prize and exactly one true label.

Make it easier

Use a three-row table with an adult reading the labels aloud.

Take it further

Change the condition to exactly two true labels and explain why there are two possible locations.

Another day, away from the screen

Invent a two-box puzzle and check whether its clues produce zero, one or several solutions.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. The shortest delivery route

A little hint

A to B to C costs 3, less than the direct A to C cost of 5.

Answer / what to notice

The shortest route is A-B-C-D-E with cost $2+1+1+3=7$.

The cheapest costs from A are B=2, C=3, D=4 and E=7; the diagram's drawn distances are irrelevant.

Make it easier

Compare only A-B-D-E and A-B-C-D-E.

Take it further

Close the C-D link and find the equally best routes. Both A-B-D-E and A-B-C-E cost 9.

Another day, away from the screen

Design a five-stop network with two equally cheap routes.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Crack the four-digit order

A little hint

Arrange the three pieces 13, 4 and 2; then apply the other two clues.

Answer / what to notice

The code is 4213, read left to right: four, two, one, three.

With 4 before 2, the possible block orders are 13-4-2, 4-13-2 and 4-2-13; only the last avoids ending in 2.

Make it easier

Provide the three possible block orders to compare.

Take it further

Remove one clue and count how many codes become possible.

Another day, away from the screen

Write an ordering puzzle with four animal names and verify its solution count.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Thirteen-token strategy

A little hint

If the other player takes 1, take 3; if they take 2, take 2; if they take 3, take 1.

Answer / what to notice

The first player can take 1, leaving 12, then reply so each pair removes 4, leaving 8, 4 and finally 0.

Keeping the opponent's starting total at a multiple of 4 forces the final winning reply when the rule is followed.

Make it easier

Start with 5 tokens and test the same first move.

Take it further

Change the maximum take to 4 and investigate a useful starting total.

Another day, away from the screen

Teach the strategy to a partner, then cooperate to explain why it works.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

6. Every fruit label is wrong

A little hint

The Mixed-labelled box cannot really be mixed.

Answer / what to notice

Mixed-labelled: apples only. Bananas-labelled: mixed. Apples-labelled: bananas only.

The sample fixes apples; the Bananas label must still be wrong, forcing mixed there and bananas in the last box.

Make it easier

Resolve the Mixed-labelled box first.

Take it further

Repeat the reasoning if the sample was a banana.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

7. Two speakers, fixed habits

A little hint

Test A truthful, then A lying.

Answer / what to notice

A lies; B tells truth.

If A were truthful, B would lie but B's sentence would be true, a contradiction. The opposite assignment is consistent.

Make it easier

Make a table of the two possible assignments suggested by A.

Take it further

Change B's sentence to "We both lie" and test whether a solution remains.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

8. The earliest project finish

A little hint

Assembly waits for the slower first task.

Answer / what to notice

Minute 12.

Both first tasks are finished at minute 3; assembly ends at 7 and drying at 12.

Make it easier

Draw a timeline for each person.

Take it further

Compare with one person doing cutting and folding sequentially: 14 minutes.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

9. Can three lights all turn on?

A little hint

Track whether the number on is even or odd.

Answer / what to notice

No.

Flipping two lights changes the count by plus 2, zero, or minus 2. It stays even, while three is odd.

Make it easier

List the states after one press.

Take it further

Add a button that flips only light 1; find a way to light all three.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

10. A four-by-four number square

A little hint

Row 1 needs 2 and 3; column 2 already has 3.

Answer / what to notice

Rows: 1 2 3 4; 3 4 1 2; 2 1 4 3; 4 3 2 1.

Each placement must satisfy its row, column and block; checking all three rules gives this unique grid.

Make it easier

Fill row 1 as 1 2 3 4 before continuing.

Take it further

Explain a chain of forced placements without guessing.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

11. Walk every bridge once

A little hint

The D bridge must be at an end of the walk.

Answer / what to notice

D-A-B-C-A works; D-A-C-B-A also works.

Each listed bridge is used once; revisiting a place is allowed.

Make it easier

Cross off each bridge after tracing it.

Take it further

Explain why starting at B cannot use every bridge exactly once.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

12. Three-digit clue laboratory

A little hint

Test sets of three digits before their order.

Answer / what to notice

415, read four, one, five.

It contains only 1 from 123 in a new place; 4 and 5 from 452 with only 4 placed right; 1 and 4 from 314 with only 1 placed right.

Make it easier

Reveal that the code contains 1,4,5 and solve their order.

Take it further

Show why each of the other five orders of those digits fails.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

13. The club overlap

A little hint

Adding the club totals counts some children twice.

Answer / what to notice

Three.

$8+7=15$ memberships for 12 children; the extra 3 are the second memberships of those in both.

Make it easier

Draw two overlapping circles and twelve marks.

Take it further

Find art-only and science-only totals: five and four.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

14. What follows from some?

A little hint

Follow the red pieces that are circles; other red pieces are unspecified.

Answer / what to notice

Some must be smooth; every red piece need not be.

The red circles inherit smoothness; the clues say nothing about red non-circles.

Make it easier

Draw one red smooth circle and one red rough square.

Take it further

Create another example that satisfies both clues but refutes “all red pieces are smooth”.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

15. Rotate the round-table plan

A little hint

Relative positions can stay the same while every seat changes.

Answer / what to notice

C remains opposite A.

A and C each move the same distance around the circle, keeping two seat-steps between them.

Make it easier

Draw old and new positions in separate circles.

Take it further

Move only A and B by swapping their seats; identify A's new opposite.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

16. Nine choices and yes-or-no clues

A little hint

Count the leaves of a three-level yes/no tree.

Answer / what to notice

No. Three questions allow at most eight answer patterns; four can distinguish nine.

Each yes/no answer can at most double the distinguishable outcomes: 2,4,8,16.

Make it easier

Build the tree for two questions first.

Take it further

Design a four-question strategy by splitting choices as evenly as possible.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

17. How many knockout matches?

A little hint

Count eliminated teams instead of drawing every possible bracket.

Answer / what to notice

Seven matches.

Seven of eight teams must be eliminated, one per match.

Make it easier

Use four teams and remove one mark per match.

Take it further

Generalise to any positive number of teams and explain why the result is teams minus one.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

18. A five-square picture code

A little hint

After the first empty cell, only four cells remain.

Answer / what to notice

Cells 2,3 and 5.

The two-cell run, separating gap and one-cell run use all four remaining cells.

Make it easier

Mark cell 1 empty and reserve cell 4 as the separator.

Take it further

Remove the first-cell clue and list every valid arrangement.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

19. The damaged checkerboard

A little hint

Every domino covers one square of each colour.

Answer / what to notice

No.

Originally there are 8 of each colour. Removing two of one colour leaves 6 and 8, but seven dominoes require 7 and 7.

Make it easier

Count and write both remaining colour totals.

Take it further

Explain why equal colour counts would be necessary but do not alone prove every board tileable.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

20. Everyone meets once

A little hint

Counting five partners for each person counts every pair twice.

Answer / what to notice

Fifteen conversations.

Six times five gives 30 partner counts; divide by two because A-B and B-A are the same pair.

Make it easier

List pairs for three people first.

Take it further

Add a seventh person and explain why six new conversations appear.

Another day, away from the screen

On another day, draw your own version of this challenge and explain which details must stay the same for the answer to work.